Laboratory Diagnosis of Haemoglobin Disorders

1. Thalassaemia
2. Haemoglobinopathy

- Laboratory tests and their principles
- Summary of laboratory approach

Why Perform Haemoglobin Study?

- Differential diagnosis of anaemia in patients to ensure proper treatment
- Detection of carrier state in order to provide genetic counselling
- Genotyping in prenatal diagnosis

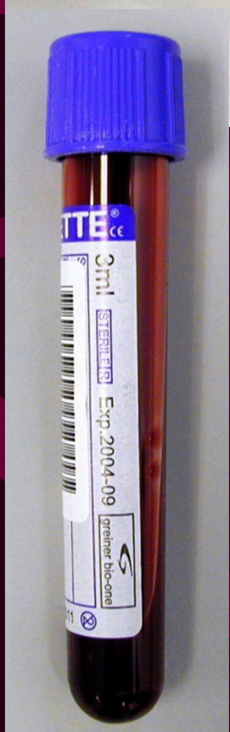
Epidemiology of Thalassaemia and Haemoglobinopathy in HK Chinese

α thal	β thal	Hb E	α thal
5%	3%	0.3%	3%
low MCV			normal MCV

Laboratory Diagnosis of Thalassaemia

- Triggers

  Low MCV +/- Clinical features
  (pallor
  splenomegaly
  failure to thrive)



Collect Date	:	05/04/03
Collect Time	:	--:--
Receive Date	:	05/04/03
Receive Time	:	18:00
Request No.	:	HN021608
Urgency	:	URGENT

Complete Blood Count

WBC	8.7
RBC	2.96
HGB	8.3
HCT	0.248
MCV	83.8
MCH	27.9
MCHC	33.3
RDW	13.5
PLT	218
MPV	9.8
Film Review	N

Clinical Classification of Anaemia

- By red cell size (Mean cell volume MCV)

Microcytic

Iron deficiency
Thalassaemia

Normocytic

Haemolysis
Anaemia of chronic disease
Renal failure

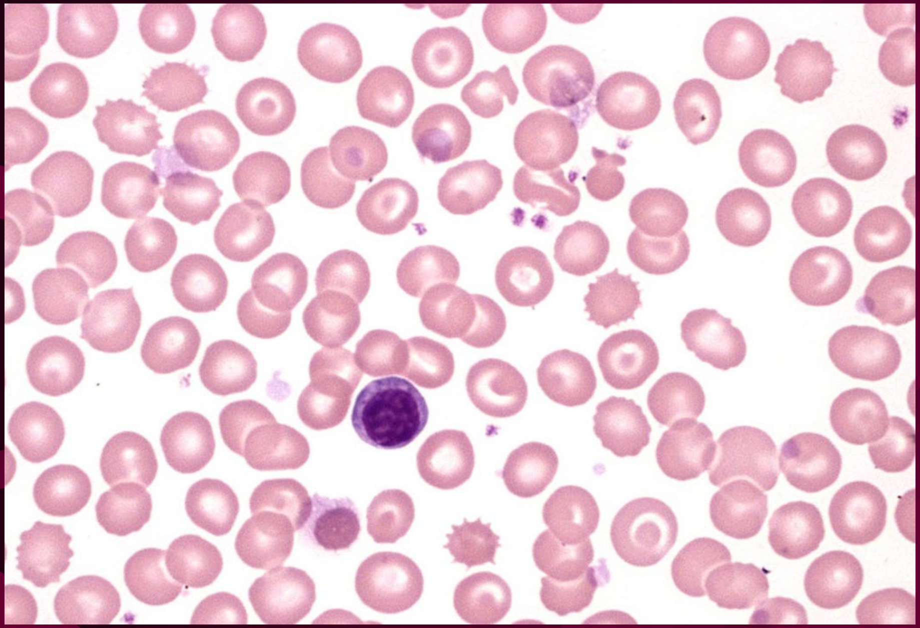
Macrocytic

Megaloblastic anaemia
Aplastic anaemia
Myelodysplasia
Liver disease

Mild Thalassaemia & Iron Deficiency Anaemia

	Thalassaemia trait	Iron deficiency
Hb	normal/slightly low	can be very low
RBC	high	low
MCV	low	low
MCH	low	low
RDW	normal/slightly high	can be very high

Blood Smear



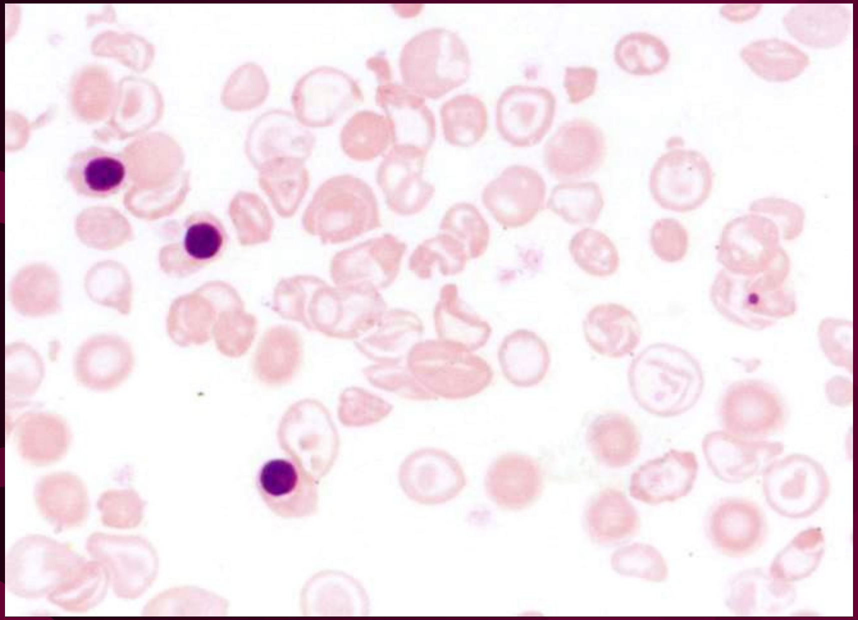
β thalassaemia trait

Iron deficiency anaemia

Severe Thalassaemia & Iron Deficiency Anaemia

	Severe thalassaemia	Iron deficiency
Hb	can be very low	can be very low
RBC	low	low
MCV	low	low
MCH	low	low
RDW	can be very high	can be very high

Blood Smear



Thalassaemia major

Iron deficiency anaemia

Haemoglobin Structure

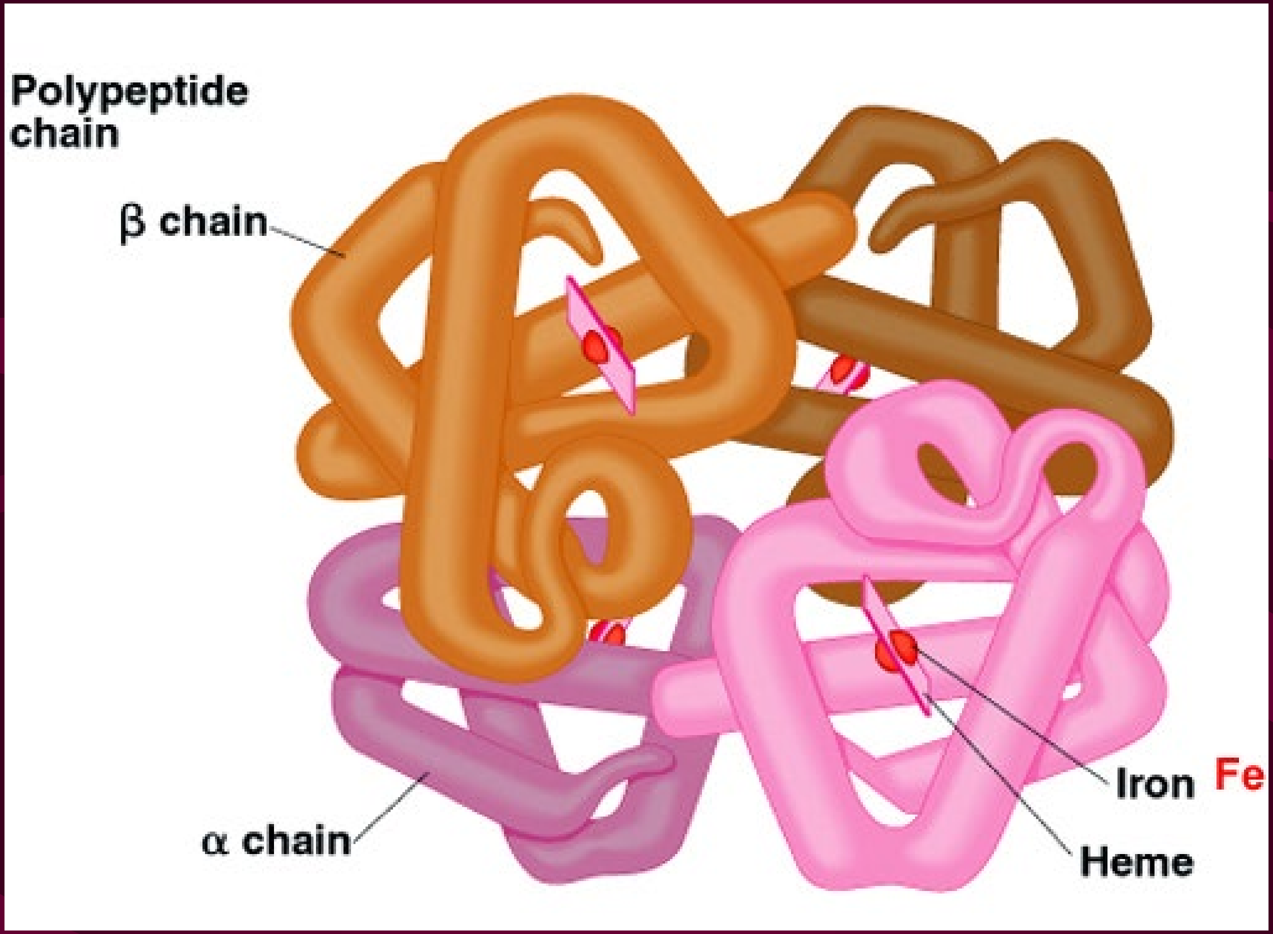
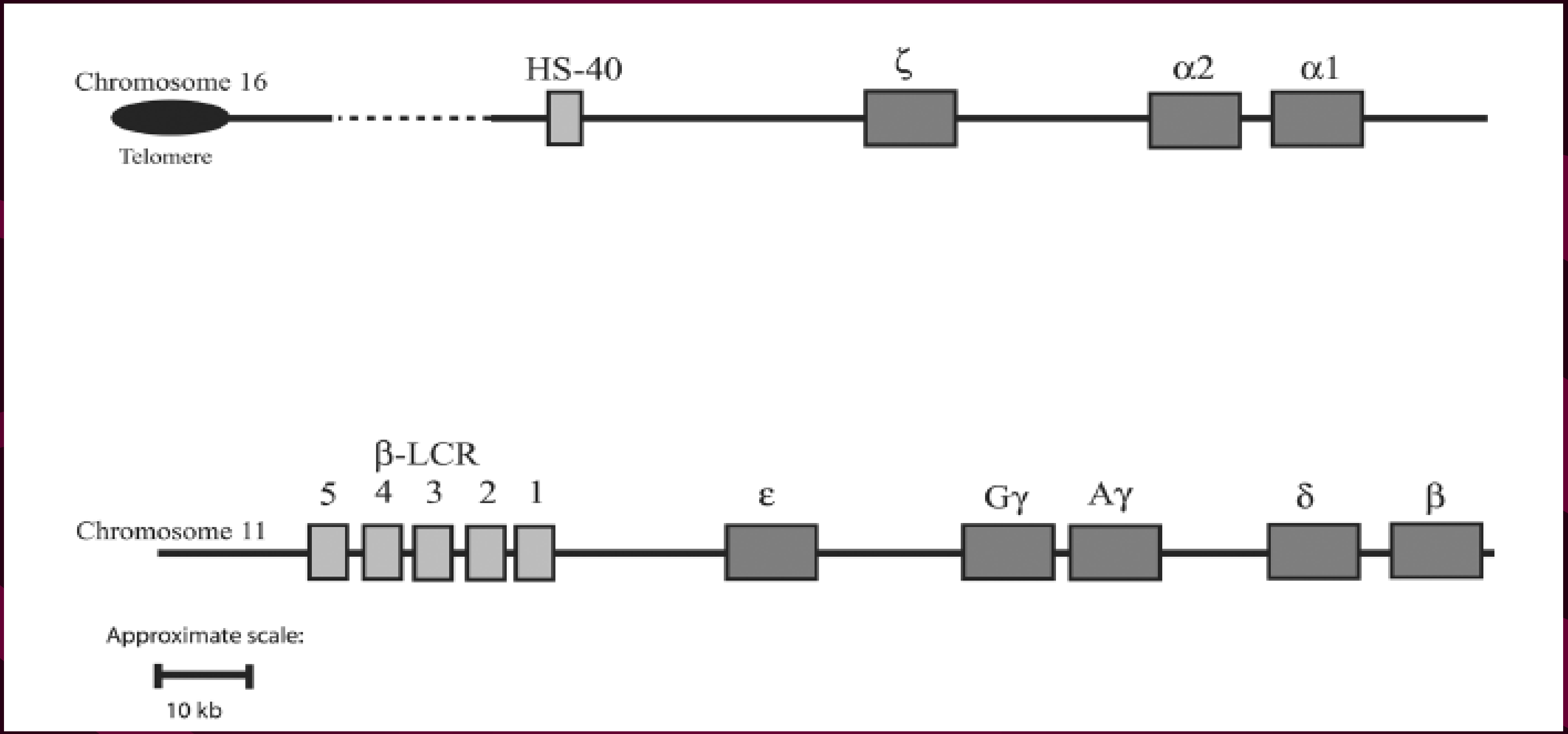
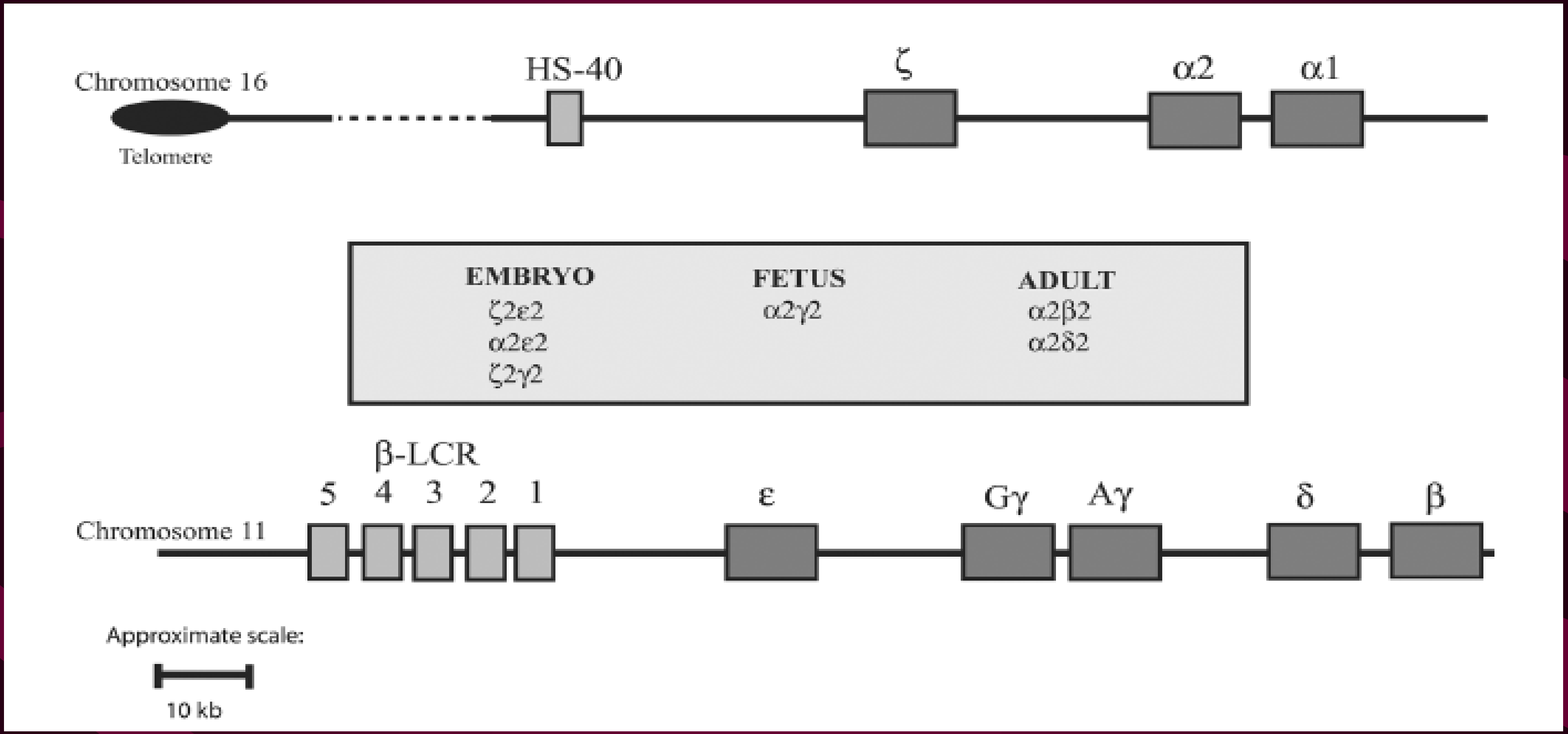


Image courtesy of the University of Miami Biology

Globin Gene Families



Haemoglobin Composition



Haemoglobin Composition in Adult

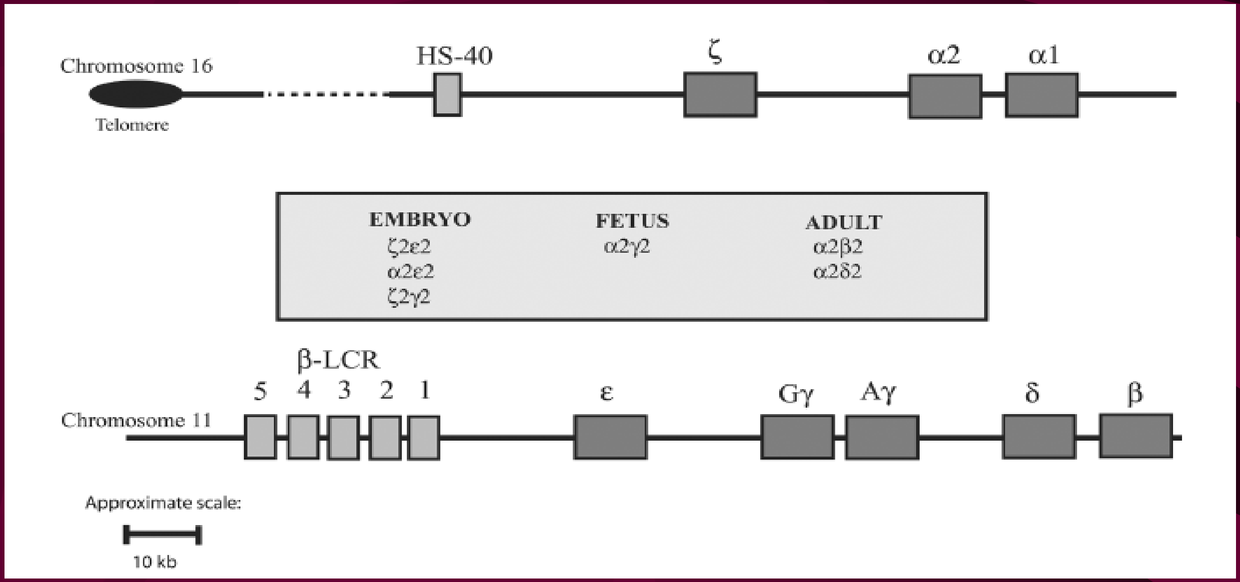
Hb A	$(\alpha_2\beta_2)$	>95 %
Hb A_2	$(\alpha_2\delta_2)$	1–3 %
Hb F	$(\alpha_2\gamma_2)$	<1 %

Thalassaemia

- Globin chain imbalance Hb A ($\alpha_2\beta_2$)

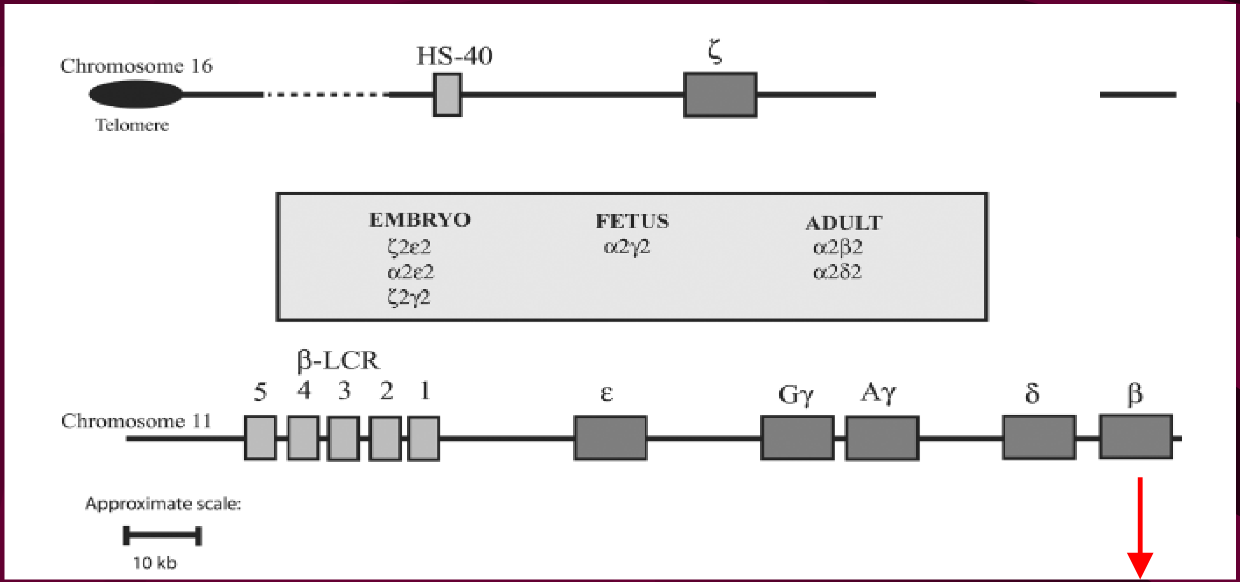
$$\alpha_{\beta} \qquad {}_{\alpha}\beta$$

Excess β Chain in α Thalassaemia

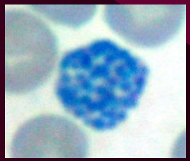


Excess β Chain in α Thalassaemia

$_{\alpha}\beta$



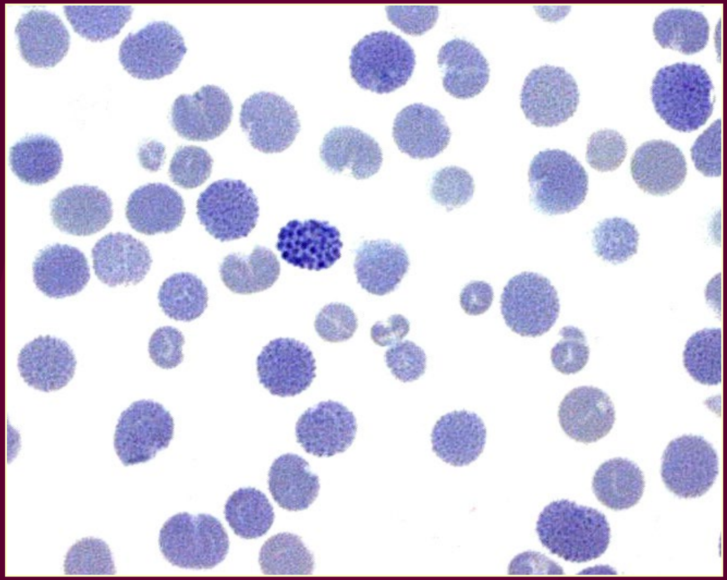
β_4 tetramer

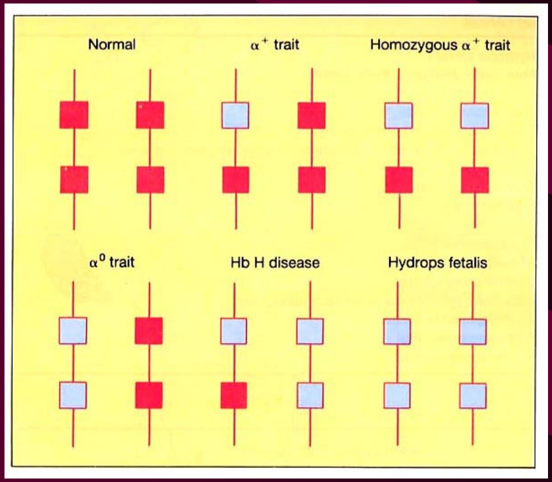


Excess β Chain in α Thalassaemia

- Supravital Staining for Hb H (β_4) Inclusion Bodies

α thal trait

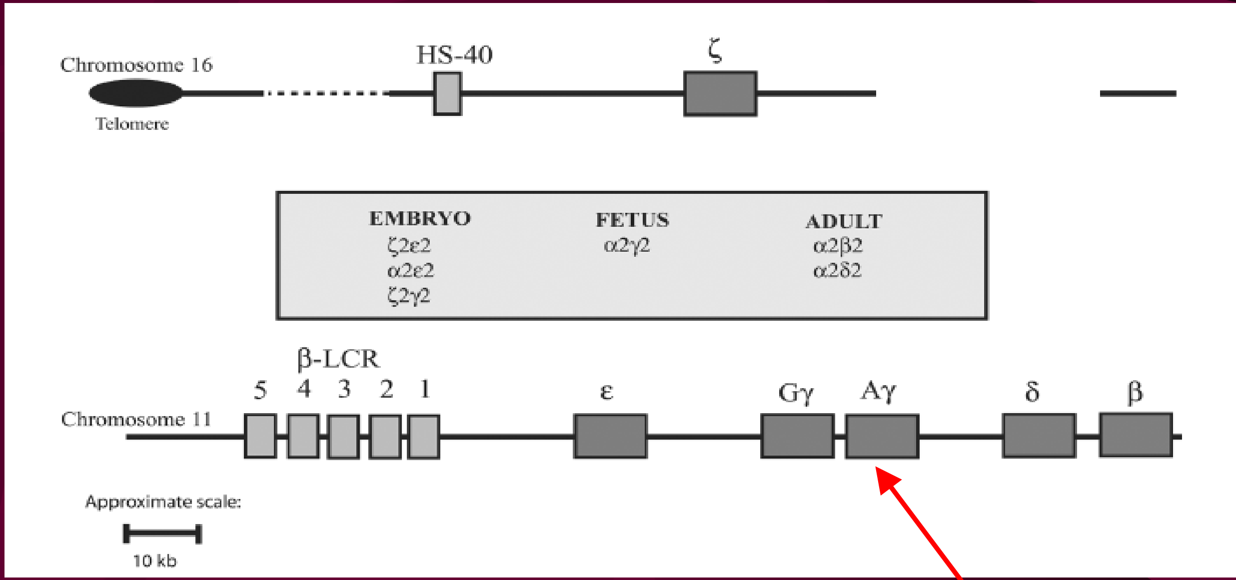




α thal intermedia / Hb H disease

Newer Method to Detect α Thalassaemia

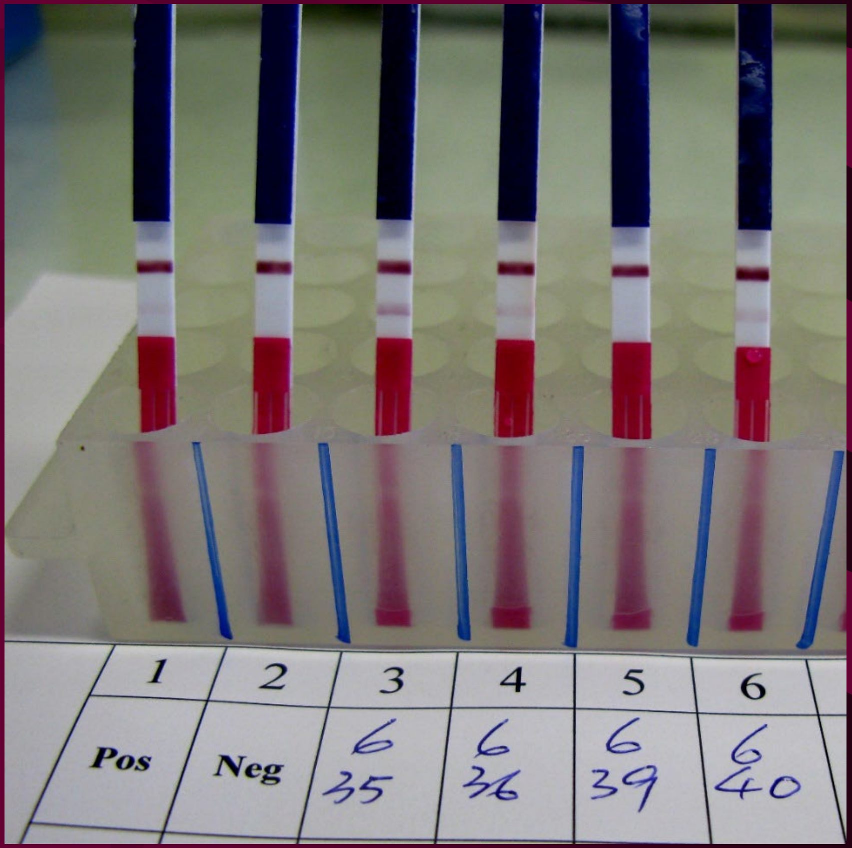
$\alpha\beta$



γ_4 tetramer

Excess γ Chain in α Thalassaemia

- Immunochromatographic strip test for Hb Bart's (γ_4)



Example of Report

(WARD ENQUIRY)

Hospital Authority
Hong Kong Children's Hospital

Urgent

Haematology Laboratory

Lab No: 21H0000302
Name:

HKID No:
Hosp No: NNC 2010025H
Location: HCH/PAE/%NNC
Sex/Age: M/1Y
Req. Loc.: HCH/PAE/NNC
Doctor:

Bed:
DOB: 05/05/2020

Date Collected: 14/05/2021 14:57
Date Registered: 14/05/2021 15:26

H

HAEMOGLOBIN ANALYSIS

	Result	Unit	Reference Intervals
HGB	11.6	g/dL	10.1 - 12.5
RBC	6.26 H	x10^12/L	4.03 - 5.07
MCV	57.8 L	fL	69.5 - 81.7
MCH	18.5 L	pg	22.7 - 27.2
RDW	16.0 H	%	12.9 - 15.6

Haemoglobin Pattern :-

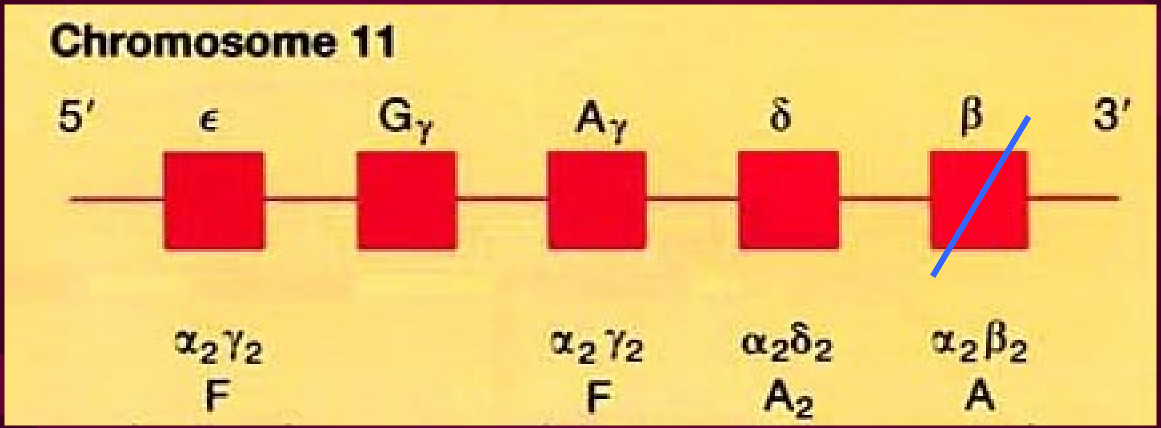
HbF	1.3	%
HbA2	2.9	%
HbH inclusion bodies	occasionally seen	

Signout Comment Findings are suggestive of alpha thalassaemia trait.

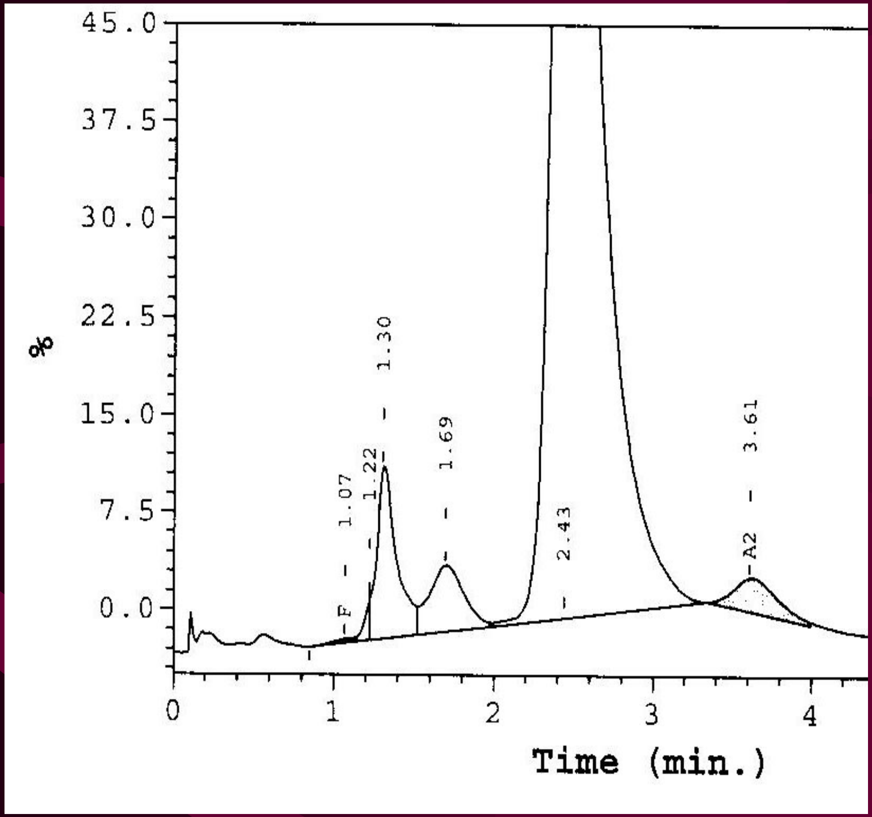
HA

Change of Haemoglobin Composition in β Thalassaemia

- Increased δ and γ chain production (increased Hb A_2 and Hb F) with decreased β chain in β thalassaemia

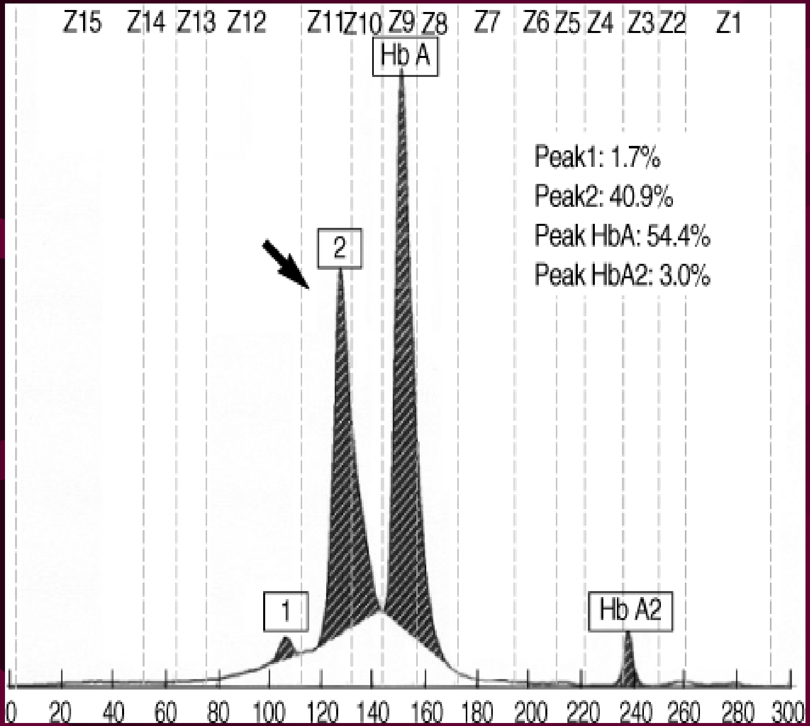


Quantitation of Hb A_2 and Hb F



High Performance Liquid Chromatography

Quantitation of Hb A_2 and Hb F



Capillary Electrophoresis

Example of Report

(WARD ENQUIRY)

Hospital Authority
Hong Kong Children's Hospital

Haematology Laboratory

Lab No: 21H0000601
Name:

HKID No:
Hosp No: ONC 1910469E
Location: HCH/ONC/%ONC
Sex/Age: M/12Y
Req. Loc.: HCH/ONC/ONC1
Doctor:

Bed:
DOB: 17/05/2009

Date Collected: 23/08/2021 13:59
Date Registered: 23/08/2021 14:13

H

HAEMOGLOBIN ANALYSIS

			Reference Intervals
HGB	11.4	g/dL	11.0 - 14.5
RBC	5.72 H	x10^12/L	4.03 - 5.29
MCV	61.4 L	fL	76.7 - 89.2
MCH	19.9 L	pg	25.2 - 30.2
RDW	16.3 H	%	12.4 - 14.5

Haemoglobin Pattern :-

HbF	2.4	%
HbA2	5.8	%
HbH inclusion bodies	not seen	

Signout Comment Consistent with beta thalassaemia trait

Example of Report

(WARD ENQUIRY)

Hospital Authority
Hong Kong Children's Hospital

Urgent

Haematology Laboratory

Lab No: 21H0000382
Name

HKID No:
Hosp No: HN21007290C
Location: HCH/SUGE/6N
Sex/Age: M/5Y
Req. Loc.: HCH/SUGE/6N
Doctor:

Bed: DISCHARGED
DOB: 11/12/2015

Date Collected: 10/06/2021 12:31
Date Registered: 10/06/2021 14:48

H

HAEMOGLOBIN ANALYSIS

			Reference Intervals
HGB	8.6 L	g/dL	10.2 - 12.7
RBC	3.24 L	x10^12/L	3.89 - 4.97
MCV	76.9	fL	71.3 - 84.0
MCH	26.5	pg	23.7 - 28.3
RDW	12.3 L	%	12.5 - 14.9

Haemoglobin Pattern :-

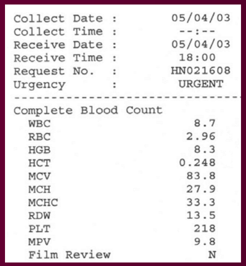
HbA	present	
HbF	<1.0	%
HbA2	2.8	%
HbH inclusion bodies	not seen	

Signout Comment No evidence of thalassaemia

HAE

Patient Sample

MCV Screening



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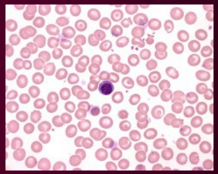
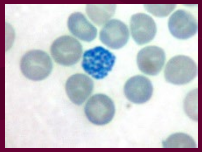
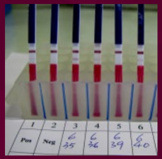


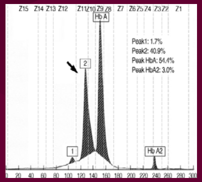
Supravital Staining / IC Strip Test

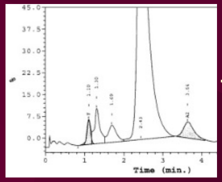




α thalassaemia

HPLC / Capillary Electrophoresis





β thalassaemia

Approach for Laboratory Diagnosis of Thalassaemia

Laboratory Diagnosis of Haemoglobinopathy

- Triggers

Clinical features	+	Laboratory findings
(pallor, jaundice splenomegaly)		(haemolysis)
(plethora)		(erythrocytosis)
(cyanosis)		(methHb, low SaO_2)

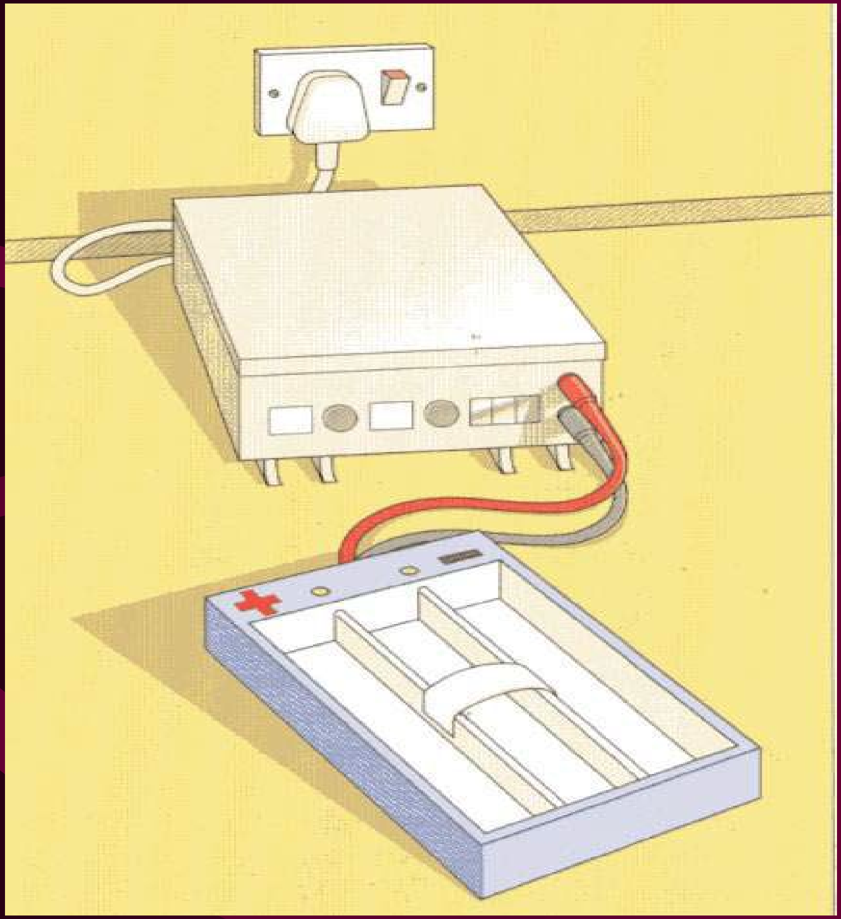
Relatively Common Hb Variants in HK

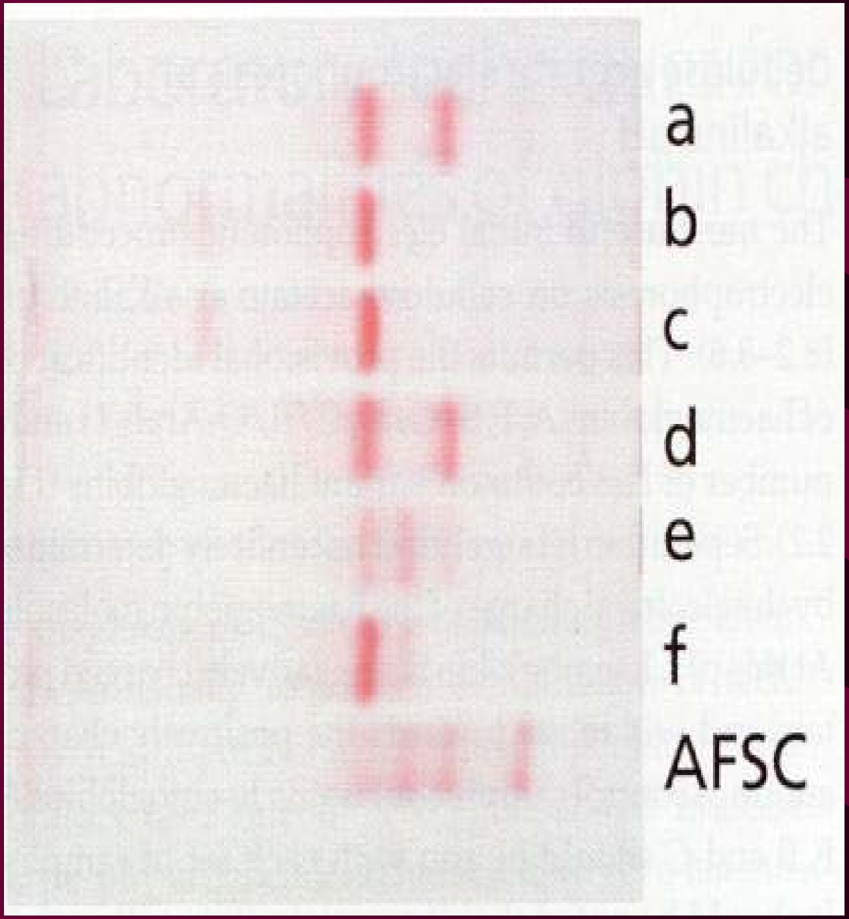
- Hb E (β26 Glu>Lys)
  - reduced production due to creation of a new alternative splice site, normal oxygen affinity, mildly unstable
- Hb Constant Spring (α2 142, Stop>Gln)
  - markedly reduced production due to unstable mRNA

Detection of Haemoglobin Variants

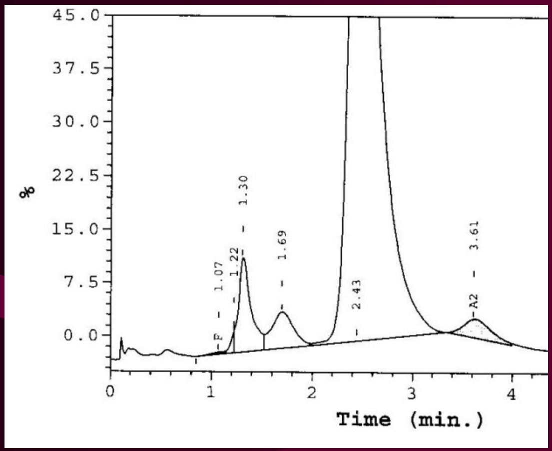
- Change in overall charge
- Change in solubility
- Change in stability
- Change in oxygen affinity
  - Some variants have changes in multiple properties

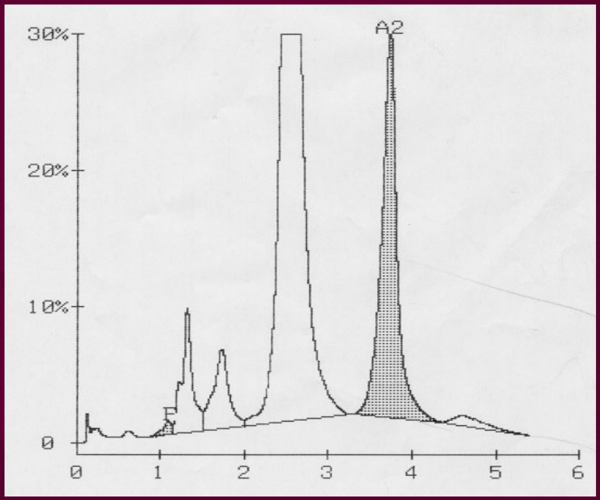
Altered Electrophoretic Mobility



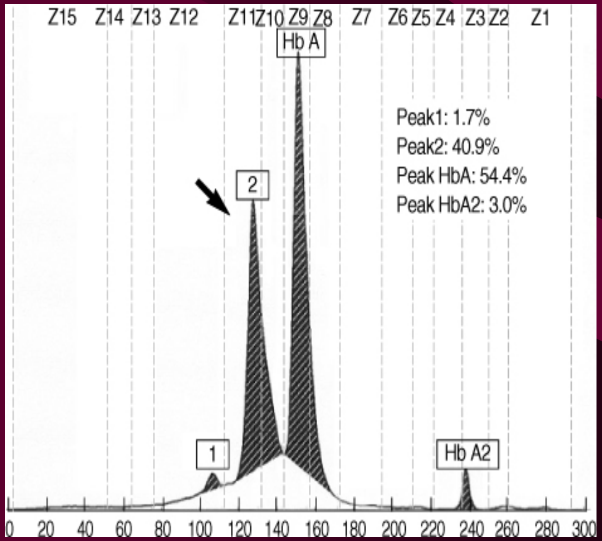


HPLC Detection of Hb Variants





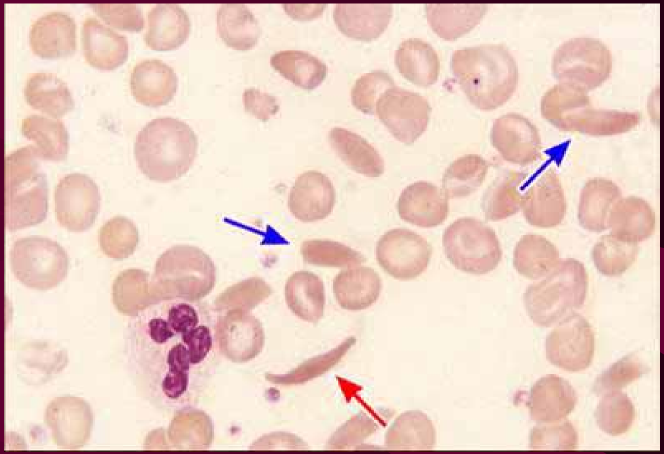
HPLC



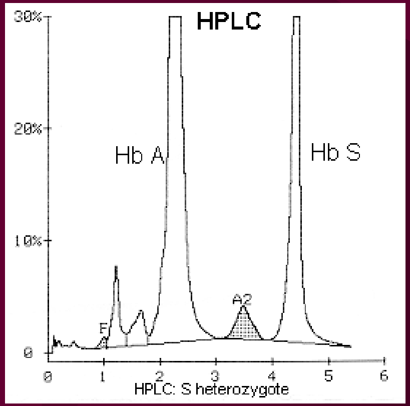
Capillary Electrophoresis

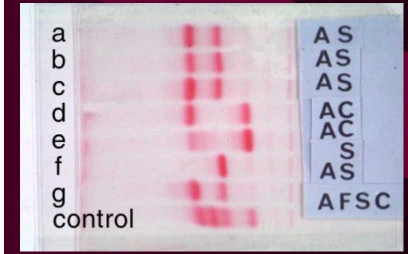
Altered Solubility - Detection of Hb S

- Peripheral blood smear
- Hb S solubility test



- HPLC / Electrophoresis

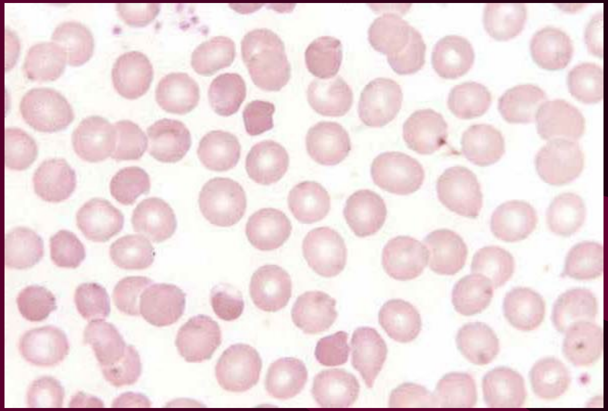


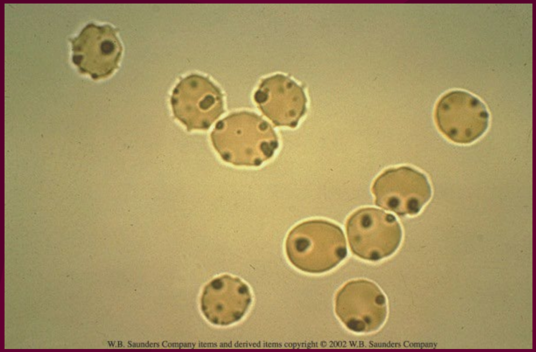


Interactive Haematology ImageBank B. Bain

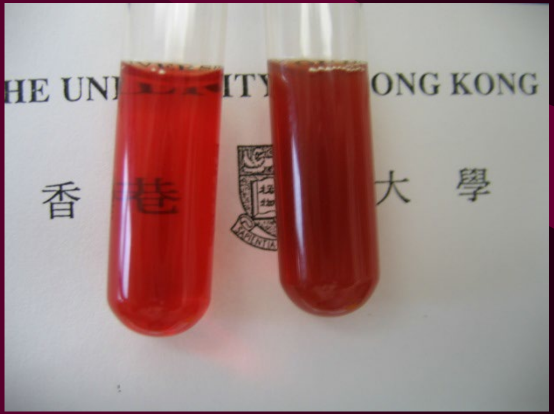
Altered Stability - Detection of Unstable Hb

- Precipitation of unstable Hb
- Peripheral blood smear
  - Irregularly contracted cells
  - Heinz bodies



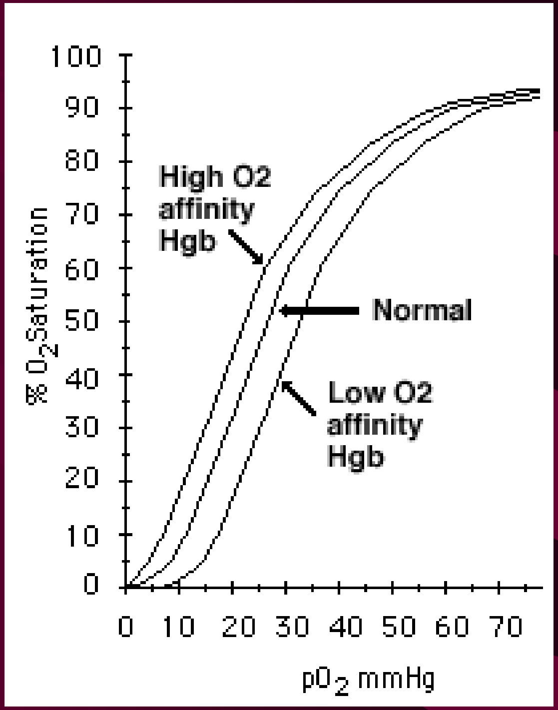


- Heat stability test
- Isopropanol stability test

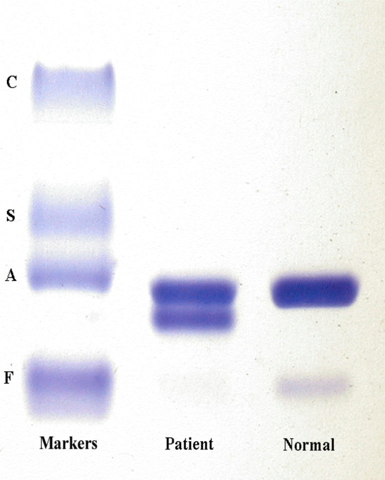


Altered O_2 Binding - Detection of Hb with Altered O_2 Affinity

- Oxygen saturation study
  - P_{50} calculator by PO_2 (from blood gas) and SO_2 (from co-oximetry) https://www.medsci.org/v04/p0232/ijmsv04p0232s1.xls



- Gel electrophoresis



Approach for Laboratory Diagnosis of Hb Variants

Patient Sample

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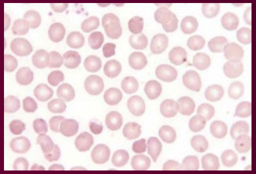
Peripheral Blood Smear →



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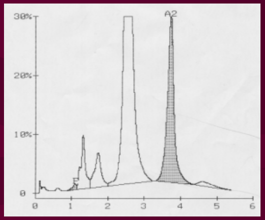
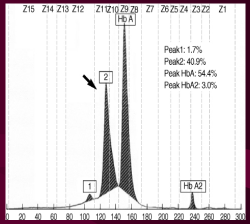
HPLC / Capillary Electrophoresis →





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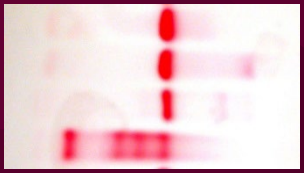
Electrophoresis →



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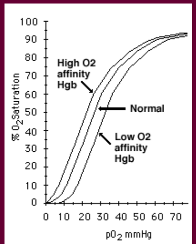
Hb Solubility test

Hb Stability Test

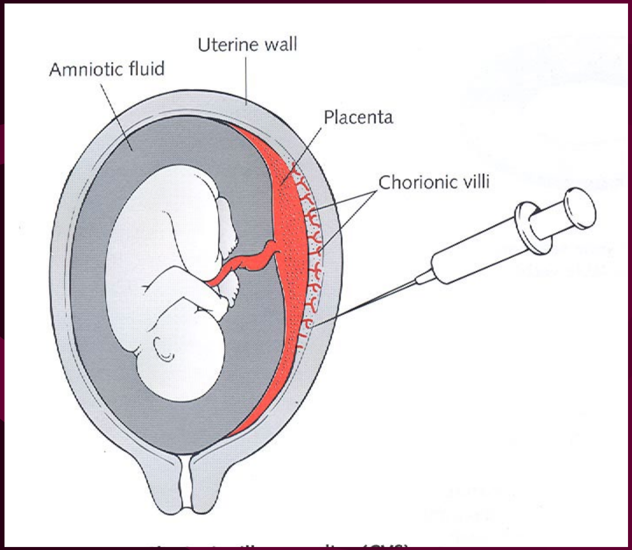
O_2 Saturation Study

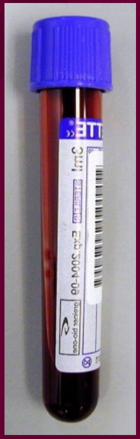


Molecular Study of Globin Diseases

- To investigate patients with atypical phenotypes
- To confirm identity of Hb variant
- Use in prenatal diagnosis

Prenatal Diagnosis of Thalassaemia





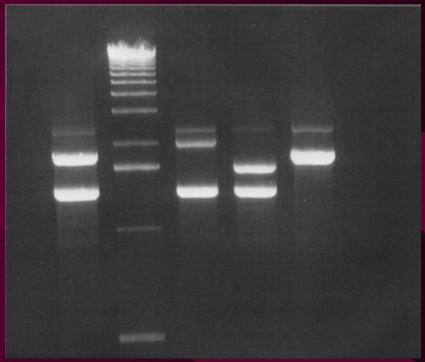
- Preimplantation genetic diagnosis
- Chorionic villous biopsy
- Aminocentesis
- Fetal DNA in maternal plasma

Genetic Defects of Globin Genes in Hong Kong Chinese (Low MCV)

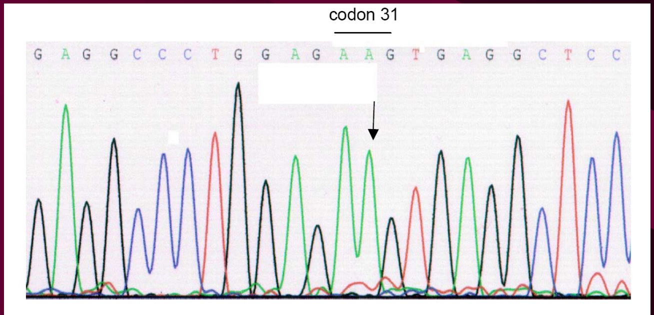
α-thalassaemia	
(--^{SEA}) α-thalassaemia deletion	90%
β-thalassaemia point mutations	
codons 41-42 (-CTTT) β^0	46%
IVSII-654 (C$\rightarrow$T) β^0	28%
nt -28 (A$\rightarrow$G) β^+	13%
codon 17 (A$\rightarrow$T) β^0	6%

Genotypic Analysis of Globin

- Quick PCR-based methods for common mutations

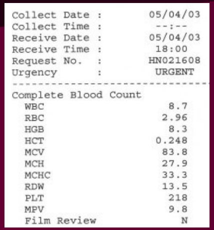


- Direct nucleotide sequencing for uncommon/novel mutations



Patient Sample

MCV, Smear examination



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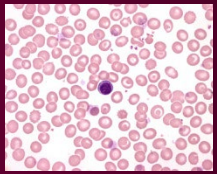
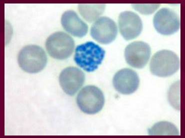
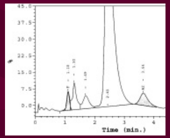
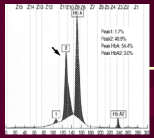


Supravital staining, HPLC,
Capillary Electrophoresis

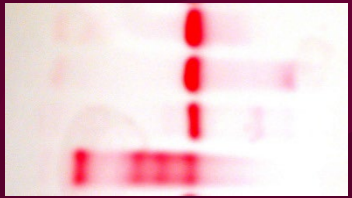


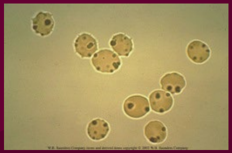




Thalassaemia

Electrophoresis,
Special tests

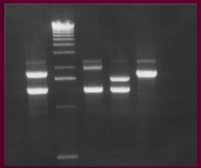


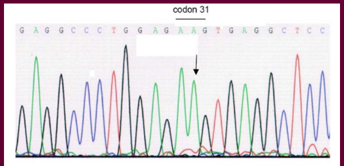


Haemoglobino
-pathy

Special indications

Molecular Analysis





Exact mutation

References

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- So, CCJ, Ma ESK (2022). Hemoglobin and hemoglobinopathies. In N Rifai, RWK Chiu, I Young, CD Burnham & CT Wittwer (Eds.). *Tietz Textbook of Laboratory Medicine* (7th ed, pp. 3311-3347). Elsevier.

TELL US WHAT

SC Whole Class Session

YOU THINK

SC
WCS

THANK YOU!

Look for today's teacher
and the others you may have missed.
(HKU Portal login required)